

TENACIOUS INTELLIGENCE CORPORATION

Internal Technical Memo — Restricted Distribution

TO: Product & Engineering Leadership
FROM: Birkity Yishak, Research Partner -- TRP1 Week 11
RE: Tenacious Judge v0.1 — SimPO LoRA Adapter Evaluation Report (Path B)
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PAGE 1 — EXECUTIVE BRIEF

Executive Summary

The Tenacious Judge — a SimPO-tuned LoRA adapter on Qwen2.5-3B-Instruct trained on 228 dimension-reasoning preference pairs derived from Tenacious-Bench v0.1 — achieves **59.4% accuracy on the 32 held-out tasks it chose to answer**, a **+8.6 percentage-point gain over the unaided base model** (50.8%; Delta B = +8.6 pp), confirming genuine preference-learning signal, especially on D2 ICP-alignment failures (14 base errors → 0 trained errors). However, the model **abstains on 50.8% of held-out tasks** (33 of 65 UNKNOWN outputs): treating abstentions as errors yields a conservative accuracy of 29.2% — **21.6 pp below the base model** — making the current adapter unsafe for unsupervised production deployment. At a total research cost of **\$0.10 USD** (dataset generation) and **\$0 compute** (Colab T4 free tier, 14 minutes), the adapter is a cost-effective proof-of-concept that warrants a structured second training iteration before any live traffic role.

1. Delta A — Gap to Perfect Ground Truth

The deterministic scorer (score_task()) defines ground truth at 100% accuracy by construction. Delta A measures how far the trained judge is from this ceiling.

Partition	n	Deterministic	Trained Judge	Delta A (trained – GT)
dev	78	100.0%	65.5% (36/55*)	–34.5 pp
held_out	65	100.0%	59.4% (19/32*)	–40.6 pp

* Accuracy computed over non-UNKNOWN responses only.

Interpretation: A –40.6 pp Delta A means the trained judge is 40 percentage points below perfect on the tasks it answers — and abstains on half the rest. This gap is not primarily caused by wrong answers; it is caused by the UNKNOWN rate. If the UNKNOWN problem is resolved, the true ceiling of the current adapter is likely in the 60–70% range on held-out, based on the dimension-level improvements observed.

2. Delta B — Lift Over Prompt-Engineered Baseline (Honest Reporting)

This is an honest negative result that must be read in full context.

The prompt-engineered baseline is the same backbone (Qwen2.5-3B-Instruct) given a structured zero-shot prompt asking for a VERDICT: PASS or REJECT. This is the fairest comparison: identical model, identical hardware, only the adapter differs.

Metric	Base Model	Trained Judge	Delta B
Accuracy (scored only)	50.8% (33/65)	59.4% (19/32)	+8.6 pp
UNKNOWN rate	0%	50.8% (33/65)	—
Conservative accuracy (UNKNOWN = wrong)	50.8%	29.2% (19/65)	−21.6 pp
D1 errors (grounding)	13	8	−5
D2 errors (ICP align)	14	0	−14

Honest assessment: The +8.6 pp headline applies only to the 49% of tasks where the model produces a parseable output. The base model, which never abstains, has a better conservative accuracy (50.8%) than the trained judge (29.2%). The adapter is not better than the base model in an end-to-end production setting without an abstention-fallback mechanism. The D2 improvement (14 → 0 errors) is a genuine learning signal and the strongest result of this training run.

3. Cost Per Task and Production Implication

Component	Cost	Per-Task Cost	Notes
Dataset generation	\$0.10 total	\$0.000389	DeepSeek V3.2 via OpenRouter, 257 tasks
SimPO training	\$0.00	\$0.00	Google Colab T4 free tier, 14 min
Deterministic scorer	~\$0.00	< \$0.001 ms	No LLM, no GPU
Base model inference	~\$0.002/task	\$0.002	Qwen2.5-3B @ T4, ~40s/task est.
Trained judge inference	~\$0.002/task	\$0.002	Same GPU cost; adapter adds <5% overhead

Production implication: The trained judge costs the same to run as the base model (both require GPU inference at ~\$0.002/task). The deterministic scorer is effectively free. The only economic argument for the trained judge over the deterministic scorer is on tasks the deterministic scorer cannot cover — specifically D2 Phase 2 semantic alignment, which requires LLM inference regardless. For D1, D3, D4, and D5 (all fully deterministic), the trained judge adds \$0.002/task with no accuracy improvement over the free scorer. The break-even point: the trained judge is worth its compute cost only if it catches at least one additional D2 error per 500 tasks compared to the base model ($\$0.002 \times 500 \text{ tasks} = \1.00 , the cost of one missed deal-quality signal).

4. Production Recommendation

Recommendation: DEPLOY WITH CAVEAT

Deploy the trained judge in a **hybrid fallback architecture**: run the trained adapter first; if output is UNKNOWN, fall back to the base model's zero-shot verdict; for D1/D3/D4/D5, always use the deterministic scorer as final arbiter. The trained adapter should only influence the final verdict on

D2 cases.

Conditions required for full (unconditional) deployment:

- UNKNOWN rate reduced to <10% on a 100-task held-out evaluation (currently 50.8%)
- Conservative accuracy (UNKNOWN=wrong) exceeds base model baseline of 50.8%
- D2 zero-error rate holds on a new evaluation batch with novel Ambiguous-segment tasks

Interventions attempted and their outcomes:

Intervention	UNKNOWN Count (dev)	Result
No intervention (original)	78/78	Complete failure
Increase max_new_tokens (128→2048)	Partial	Minor improvement
Response priming: append 'VERDICT: significant drop'	Significant drop	Partial fix
Add system prompt enforcing verdict-23/78 (dev)	23/78	Best achieved
All three combined	23/78 dev, 33/65 held-out	Current deployed state

Root cause hypothesis: 228 training pairs on a 3B-parameter model is insufficient to fully override the backbone's instruction-following distribution. The model partially learned the structured output format but defaults to extended reasoning mode on novel prompts. This is a training-data volume and format-enforcement failure, not a benchmark design flaw. It would require either (a) 1,000+ pairs with format-enforcement reward, or (b) a DPO-style fine-tuning pass specifically targeting output contract compliance.

8. Kill-Switch Trigger Condition

The following measurable conditions trigger immediate rollback of the trained adapter and reversion to base-model-only inference:

Trigger	Threshold	Rationale
UNKNOWN rate (rolling 100-task window)	<30%	Current held-out UNKNOWN = 50.8%. Threshold of 30% allows modest distribution shift.
REJECT precision (weekly 20-task human audit)	>0.85	Trained judge REJECT precision on held-out ≈ 0.73 on scored tasks. Threshold set at 0.85.
Delta B on monthly re-evaluation (trained <100)	<100	Trained judge's conservative accuracy (UNKNOWN=wrong) falls below the baseline.

Artifacts

Artifact	URL / Location
Dataset (HF)	https://huggingface.co/datasets/Birkity/tenacious_bench_v0.1
Model adapter (HF)	https://huggingface.co/Birkity/tenacious-judge
Community post	https://www.reddit.com/r/FunMachineLearning/comments/1t1mlby/
Training notebook	training_running.ipynb (repo root)
Ablation results	ablations/ablation_results.json
Evidence graph	evidence_graph.json